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# Paleomagnetism of middle–late Jurassic sediments from Poland and implications for the polarity of the geomagnetic field

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## Abstract

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Paleomagnetic data for the middle–late Jurassic sediments from nine localities of Krakow-Czestochowa Upland belonging to stable Europe, and from three valleys of the Lower Subtatric Nappe belonging to mobile Europe are presented.

The majority of the rocks investigated comprising limestones, stromatolites and radiolarites were found to be normally magnetized. Natural remanent magnetizations (NRM) of reversed polarity were found in limestones and stromatolites of middle Callovian age from the stable region and in one exposure of Oxfordian radiolarites from the mobile region. Along one of the middle Callovian sections studied the NRM direction of the specimens changed from R to N.

The mean paleomagnetic pole positions obtained for the Krakow–Czestochowa Upland (lat. 72.3° N, long. 150.4° E, $\alpha_{95} = 7.3°$) and for the Lower Subtatric Nappe (lat. 71.7 N, long. 132.2° E, $\alpha_{95} = 4.3°$) are consistent with the data for stable Europe (lat. 74.7° N, long. 147.1° E, $\alpha_{95} = 5.0°$).

The paleomagnetic results presented do not reveal any major relative rotations either within the Lower Subtatric Nappe or between this area and the Krakow–Czestochowa Upland. The reversed directions of NRM observed indicate the existence of a period of reversed field direction in the Middle Callovian and perhaps also in the Oxfordian.

## Introduction

In Poland Jurassic deposits are found in two distinct tectonic units here referred to as stable Europe and mobile Europe. The stable plate comprises all of Europe north of the Carpathian block and extends longitudinally from the Caucasus to the Pyrenees. Mobile Europe is the series of smaller plates which lay between the major plates—Europe to the north and Gondwana to the south—and which underwent relative displacements and rotations during the Mesozoic as a result of collision of the African and Eurasian plates. The northernmost region included in compressional tectonics forms part of southern Poland. It embraces the Polish Carpathian region with the Tatra Mountains and the Pieniny Klippen belt. The suture zone of the African and Eurasian plates is probably located slightly north of the Pieniny Klippen Belt, under the Carpathian flysch layer. Figure 1A shows the location of Jurassic sediments in Poland.

The main purpose of this study was to determine the mean paleomagnetic poles for middle–late Jurassic time for that part of Poland which belongs to stable Europe, e.g., the Krakow–Czestochowa Upland. Hitherto the only reliable paleomagnetic data for the Jurassic for

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stable Europe was that determined from samples from southern Germany (Heller, 1977, 1978), England (Hijab and Tarling, 1982), the French–Swiss Jura (Johnson et al., 1984), and the Russian Platform (Khramov, 1984).

In addition, paleomagnetic studies of sediments from the Lower Subtatric Nappe in the Carpathians (*2* in Fig. 1A) were undertaken in order to find out (a) whether this mobile region had moved relative to stable Europe, and (b) whether movements within this region itself could be detected. Further studies of the sediments from the region of the Pieniny Klippen Belt (*3* in Fig. 1A) and Kujawy (*4* in the same Figure) are currently in progress.

In this paper we present the results obtained for the Krakow–Czestochowa Upland from the following exposures (Fig. 1B): Wielun—*W*; Lisowice—*LI*; Ogrodzieniec—*OG*; Klobuck—*K1*, *K2*, *K3*; Stare Gliny—*SG1*, *SG2*; Zalas—*Z*; Podleze—*PO1*, *PO2* (stromatolites) and *PW* (limestones); Czatkowice—*CZ*. Preliminary results have already been reported (Kruczyk et al., 1980; Kruczyk and Kądziałko-Hofmokl, 1983).

Results from three exposures from the Tatra Mountains: Chochołowska Valley—*CH*; Lejowa Valley—*L*; Filipka Valley—*F*, shown in Fig. 1C, have already been published (Kądziałko-Hofmokl et al., 1985).

Preliminary results from samples of Jurassic deposits from the Balkan region in Bulgaria (Kruczyk et al., 1986) have also been included in the summarizing table.

## Geology and Sampling

### *Krakow–Czestochowa Upland*

The Jurassic deposits occurring in the Krakow–Czestochowa Upland consist mainly of pelitic sandy limestones and sandy and spongy limestones with white or grey coloring, sometimes oolitic or nodular and, in rare occurrences, stromatolites. The sedimentation rate of the limestones was rather fast, up to 1 cm per $10^3$ years while that of stromatolites has not been defined. According to Szulczewski (pers. commun., 1985) stromatolite accumulation was much faster but it is impossible to define its rate because of the biostratigraphic gap in the time of formation of these rocks.

The sediments studied range from the early Callovian to the late Oxfordian. The Krakow–Czestochowa Upland was a marine environment as late as the Bajocian, becoming deeper in the Bathonian and shallower in the Callovian, as indicated by the presence of nodular limestones and stromatolites. The maximum transgression of the sea in Jurassic time occurred during the Oxfordian when the formation of pelitic calcareous deposits was dominant. Towards the end of the Kimmeridgian the sea became shallower and finally retreated. The Polish Jurassic sea was connected with the Tethys as well as with the Russian and the Moravian seas, so one can expect to encounter similar changing paleoenvironments throughout all these regions (Książkiewicz et al., 1965).

The deposits studied from the Krakow–Czestochowa Upland form a conformable sequence of layers dipping at 2–8° NE. In the zones of transverse faults which were rather rare and of limited extent the dip angles locally reach 15°. The faults and numerous sedimentation gaps observed in the region are related to tectonic movements during the formation of Jurassic rocks, particularly during the Callovian. However, the formations selected for our study have only been subjected to vertical displacements, since there is no geological evidence for relative rotations of the studied exposures. The rocks sampled cover an area of 150 km by 80 km.

The accurate paleontological dating of the Callovian and Oxfordian deposits makes it possible to subdivide them into early, middle and late sections. Seventy hand samples were collected from nine exposures indicated in Fig. 1B, out of which two or three slices parallel to the orientation planes were cut. The slices were cut into 560 specimens in the form of cubes for paleomagnetic investigations.

### *Lower Subtatric Nappe*

The sediments studied from the Lower Subtatric Nappe are from sections situated in three valleys (CH, L, F) on the northern slopes of the

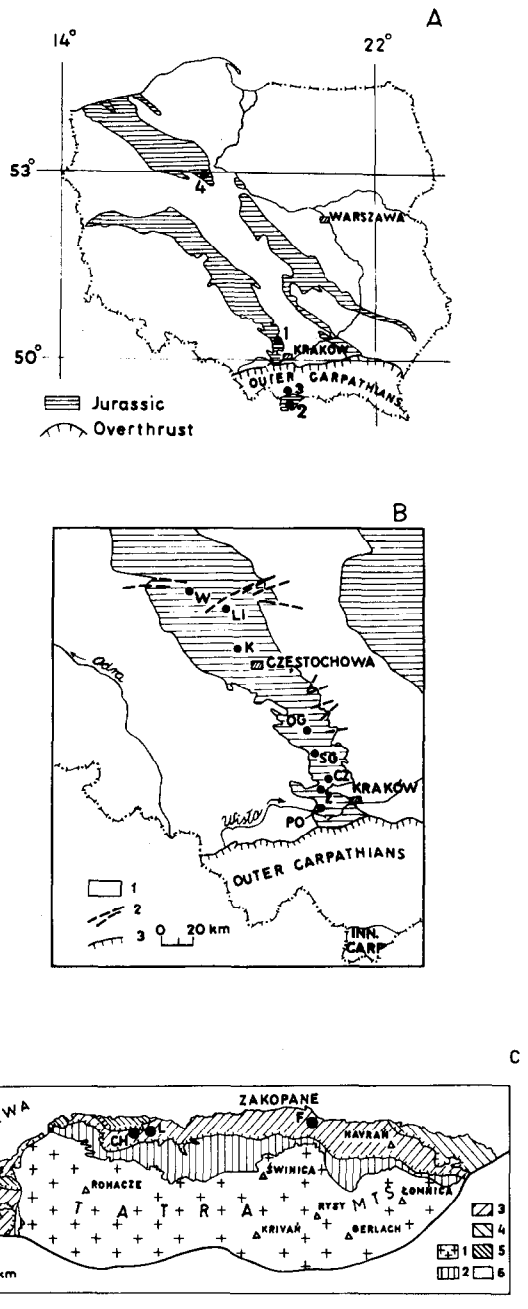



Fig. 1. Locations of the Jurassic deposits in Poland (without the Cainozoic). A. Sketch map of Poland showing the sampling areas, *1*—Krakow–Czestochowa Upland, *2*—Lower Subtatric Nappe, *3*—Pieniny Klippen Belt, *4*—Kujawy. B. Sketch map of the Krakow–Czestochowa Upland showing sampling localities referred to in the text. C. Sketch map of the Tatra Mountains showing the sampling localities of the Lower Subtatric Nappe referred to in the text. *1* = crystalline core of the Hightatric Unit, *2* = Hightatric Zone, *3* = Lower Subtatric Nappe, *4* = Bujači Partial Nappe, *5* = Middle Subtatric Nappe, *6* = Paleogene (after Lefeld, 1985).

Tatra Mountains (Fig. 1C). The Jurassic occurs here in nappes detached from the Krizna unit and displaced northward. The displacement is associated with the late Cretaceous sub-Hercynian movements during the final stage of collision of the African and Eurasian plates (Kądziałko-Hofmokl et al., 1985).

The deposits sampled were nodular limestones and red and green radiolarites. They ranged in age from the Bathonian to the Kimmeridgian. The sedimentation rate of these rocks was rather slow, about 1.5 cm/$10^4$ years. The total amount of iron compounds (as $Fe_2O_3$) in radiolarites averages from 0.63 to 0.79wt.%. The petrographic properties of the radiolarites suggest that the iron contained is of primary origin (Lefeld, 1974, Kądziałko-Hofmokl et al., 1985). The tectonic environment was more complicated than in the Krakow–Czestochowa Upland and structural corrections were more important. At exposure F the sampled radiolarites formed an overturned fold.

Seventy-seven hand samples were collected from three sections, out of which slices and then 340 specimens were cut in the form of cubes and subjected to paleomagnetic investigations.

## Method

Measurements of NRM were made with a spinner magnetometer JR3 (noise level $1 \times 10^{-5}$ A $m^{-1}$). Samples were subjected to stepwise alternating field (a.f.) demagnetization using experimental apparatus constructed in our laboratory, or heated with a 50°C increment from room temperature to 350°–550°C in an oven where the maximum magnetic field was $30 \times 10^{-3}$ A $m^{-1}$. The demagnetizing coil of the a.f. equipment was put inside a Helmholtz coil compensating the external magnetic field. A demagnetized sample rotates in a two-axis tumbler; the intensity of the demagnetizing field decreases automatically from the chosen peak value to zero. The maximum demagnetizing field is 0.1 T. Samples were demagnetized with an increment of 0.005 T or 0.01 T.

The results of demagnetization experiments were plotted on orthogonal (Zijderveld, 1967) and stereographic projections. The directions of a characteristic component of NRM (ChRM) were

obtained mainly as directions of linear segments of orthogonal plots pointing to the origin. In some cases, particularly for specimens with very low NRM intensities, the orthogonal plots failed to yield any defineable directions. In such cases the directions of ChRM were estimated using the criterion of the best grouping of NRM directions for hand samples or even whole exposures at different demagnetization levels. In a few cases the stable end point attained as a result of cleaning was taken as ChRM.

The mean directions of ChRM were calculated using Fisher statistics. For the Krakow–Czestochowa Upland we decided to calculate means for exposures rather than sites, because in most cases one exposure comprised one lithostratigraphic unit and hand samples were taken from neighboring horizons. Localities K, PO and SG where two or three different lithostratigraphic units were sampled, were divided into two (or three) exposures. The mean direction for each exposure was calculated directly from the results obtained for individual specimens. For PO1 and PO2 where R–N inversion of NRM directions was observed and where the direction of NRM changed from slice to slice, slice means were calculated (Kruczyk et al., 1980). The mean direction and pole position for the whole region were calculated from exposure means.

In the Lower Subtatric Nappe, sampling was conducted in such a way that it was possible to distinguish sites within exposures and to calculate means for each lithostratigraphic unit on the basis of the site means.

Rock magnetic experiments were conducted in order to identify the carriers of magnetic remanence. A thermomagnetic analysis was made and, for few samples, acquisition and decay curves of IRM were measured. The thermomagnetic analysis was conducted with a spinner magnetometer equipped with a nonmagnetic oven installed inside a pickup coil. Three permalloy screens ensured cancellation of the external magnetic field. The maximum temperature reached in this apparatus is 700 °C, the temperature rises linearly at a rate 20 °–30 °C/min. The experiments involved thermal demagnetization of isothermal remanence ($J_{rs}$) acquired in 1.8 T during heating in the air to 600 °C, (the isothermal remanence is denoted $J_{rs}$ because in most cases it is a saturation remanence). Thermomagnetic curves ($J_{rs}$ vs. $T$) are plotted automatically by an $XY$ recorder. From these curves, blocking temperatures ($T_b$) characterizing magnetic minerals were obtained. Experiments were conducted twice: $J_{rs}$ vs. $T$ curves obtained during the first heating show the magnetic minerals present in samples in their natural state. Curves obtained during the second heating show the composition of magnetic minerals in samples after heating in the air.

The curves of IRM acquisition and decay (IRM vs. $H$) indicate the saturation field of a sample and its bulk coercivity–both depending on the magnetic minerals present.

An analysis of polished sections and an X-ray analysis were also conducted to help in identifying magnetic minerals on a few samples of stromatolites, radiolarites and oolitic limestones.

**Rock magnetic studies**

Thermomagnetic and other analyses revealed the presence of Fe-oxides and Fe-hydroxides in the rocks investigated. The results are summarized in Table 1.

A thermomagnetic analysis was conducted for 101 specimens from the Krakow–Czestochowa Upland and for 39 from the Lower Subtatric Nappe. In exposures L, SG1, SG2 and PW the amount of magnetic minerals was so low, that it was impossible to identify carriers of magnetic remanence. We can only state that they are oxides and not hydroxides of iron. In other exposures the magnetic minerals present were positively identified. Figures 2 and 3 show thermomagnetic curves typical for the different lithological types.

The IRM acquisition and demagnetization curves were measured for four samples of stromatolites and two samples of oolitic limestones containing, according to thermoanalysis, goethite and goethite + hematite (Figs. 4A and 4B, respectively). The results indicate that neither of them attains saturation in the maximum field of experiment which was 1.8 T. The kink appearing in the field of 0.6 T on the acquisition curve for the sample containing goethite + hematite is prob-

TABLE 1

Lithology and magnetic mineralogy

| Exposure | Age | Lithology | Magnetic mineralogy * |
|---|---|---|---|
| *Krakow–Czestochowa Upland* | | | |
| OG | m. Oxfordian | spongy limestones | mgt or mgh, gth + hem |
| LI | m. Oxfordian | compact limestones | Fe-oxides? |
| PW | m. Oxfordian | pelitic limestones | Fe-oxides? |
| K1 | e. Oxfordian | spongy limestones | mgh |
| K2 | 1. Callovian | stromatolites | gth, gth + hem |
| Z | 1. Callovian | stromatolites | mgh, gth |
| SG1 | l.m. Callovian | sandy limestones | Fe-oxides? |
| CZ | m. Callovian | oolitic limestones | gth + mgh |
| PO1 | m. Callovian | stromatolites | gth, gth + hem |
| PO2 | m. Callovian | stromatolites | gth, gth + hem |
| K3 | m. Callovian | sandy limestones | gth, gth + hem |
| SG2 | e. Callovian | sandy-glauconitic limestones | Fe-oxides? |
| W | e. Callovian | sandy limestones | mgt or mgh |
| *Lower Subtatric Nappe* | | | |
| CH | Kimmeridgian | nodular limestones | mgh |
| L | Kimmeridgian | nodular limestones | mgh, hem |
| CH | Oxfordian | green radiolarites | hem, mgh |
| L | Oxfordian | red and green radiolarites | mgh, hem |
| F | Oxfordian | red radiolarites | mgt, mgh |
| L | Callovian | nodular limestones | mgh, hem |
| CH | Bajocian–Bathonian | green radiolarites | hem, mgh |
| L | Bajocian–Bathonian | green radiolarites | hem, mgh |

* mgt—magnetite; mgh—maghemite; gth—goethite; hem—hematite; Fe-oxides?—unidentified iron oxides.

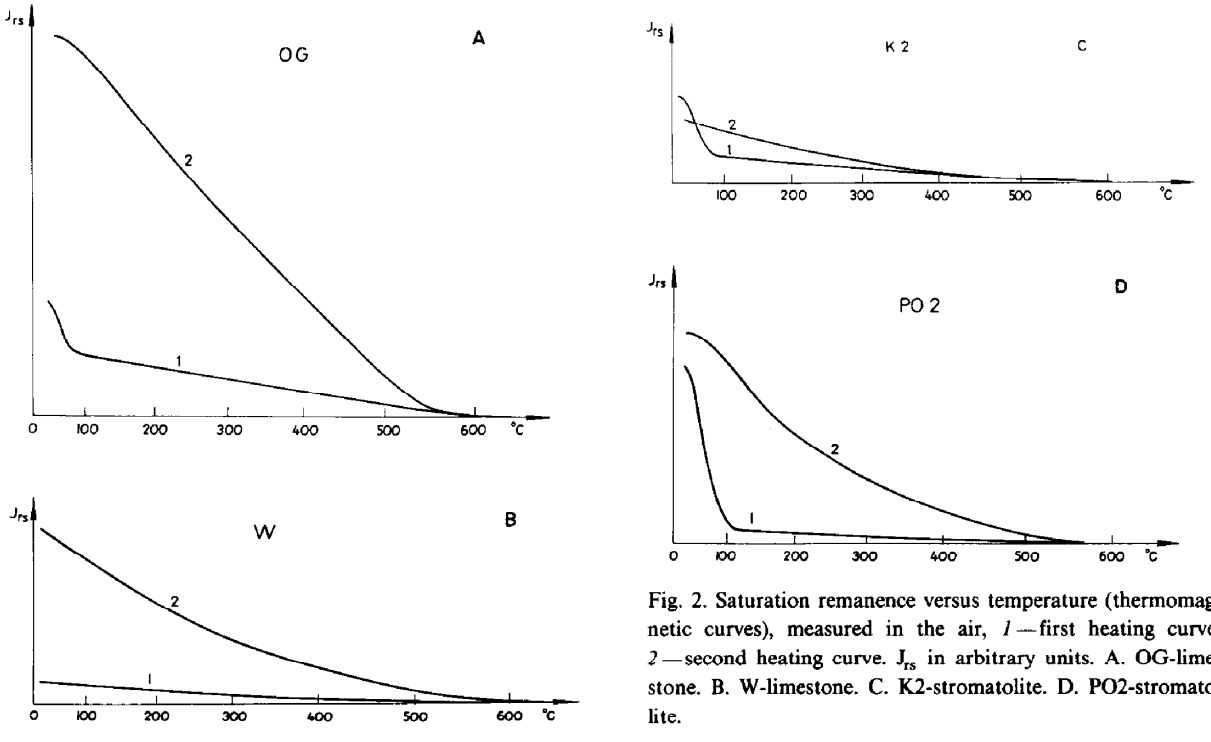



Fig. 2. Saturation remanence versus temperature (thermomagnetic curves), measured in the air, *1*—first heating curve, *2*—second heating curve. $J_{rs}$ in arbitrary units. A. OG-limestone. B. W-limestone. C. K2-stromatolite. D. PO2-stromatolite.

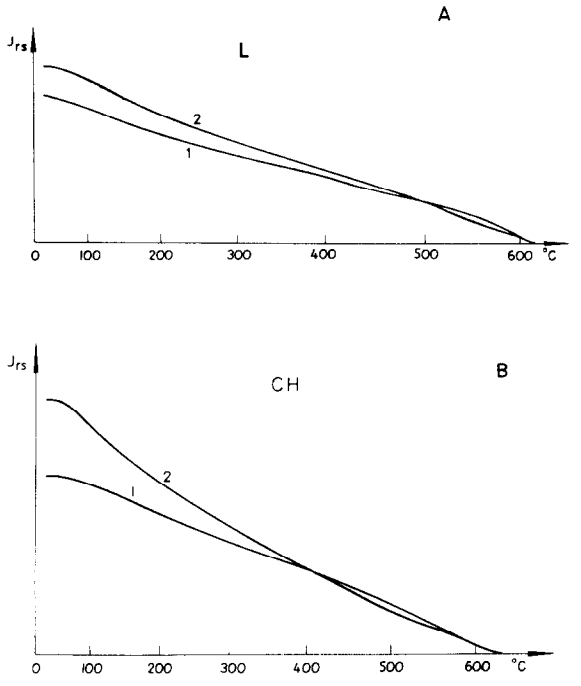



Fig. 3. Thermomagnetic curves measured in the air. A. L-limestones. B. CH-radiolarite.

ably related to hematite. The bulk coercivity of the goethite sample is higher than that of the goethite + hematite sample.

Microscopic analyses of polished sections conducted for stromatolites, radiolarites and oolitic limestones and X ray structural analysis of stromatolites confirmed the results obtained by magnetic methods. Additionally, these studies suggested that the goethite present in PO stromatolites is of primary origin and that in the oolitic limestones from CZ some magnetic oxides possibly of post-pyrite origin occur (Kruczyk et al., 1980).

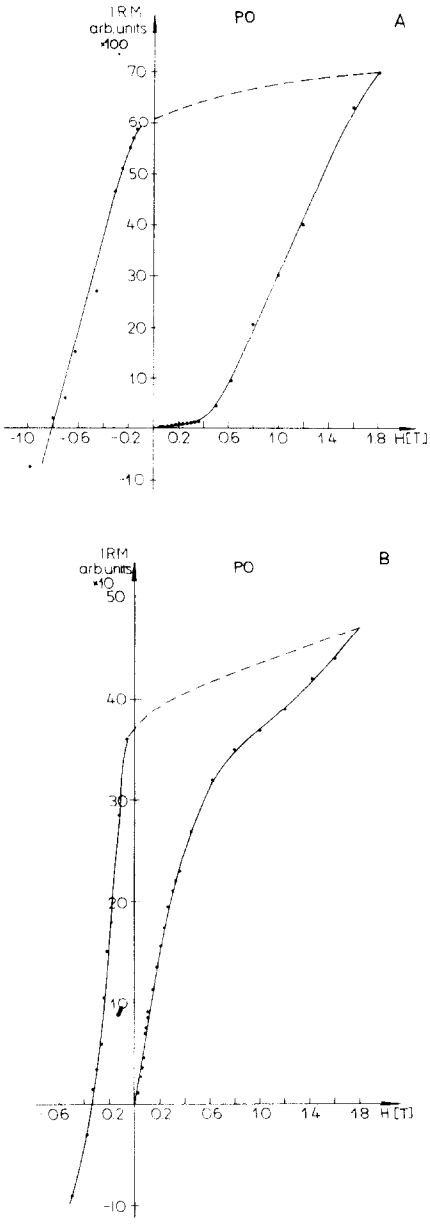



Fig. 4. IRM acquisition and decay curves. A. PO-stromatolite containing goethite. B. PO-stromatolite containing goethite + hematite.

## Paleomagnetic results

*Krakow–Czestochowa Upland*

*Oxfordian*

The Oxfordian collection comprised 170 limestone specimens from four exposures OG, LI, PW, K1 from early, middle and late subunits. The corresponding lithostratigraphic units are included in Table 1. The white and grey pelitic limestones revealed a very low NRM intensity (about $10^{-5}$ A $m^{-1}$). Figure 5A and D show typical examples of orthogonal plots. In some cases, linear segments pointing to the origin appeared on the plots. Their directions were interpreted as the directions of ChRM carried by Fe oxides and assumed to be syndepositional. In a few cases in which orthogonal plots yielded any defineable directions, the

directions of ChRM were estimated using the criterion of the best grouping of directions of NRM for hand samples at different demagnetization levels.

The mean direction of paleomagnetic field for stable Europe for the middle–late Jurassic was computed according to the reference data (Khramov, 1984 and Johnson et al., 1984; Westphal et al., 1986). The mean field direction and pole position only conforms with those expected for stable Europe in samples from exposure OG. The results obtained for the remaining three exposures do not correspond to any Jurassic or post-Jurassic European data.

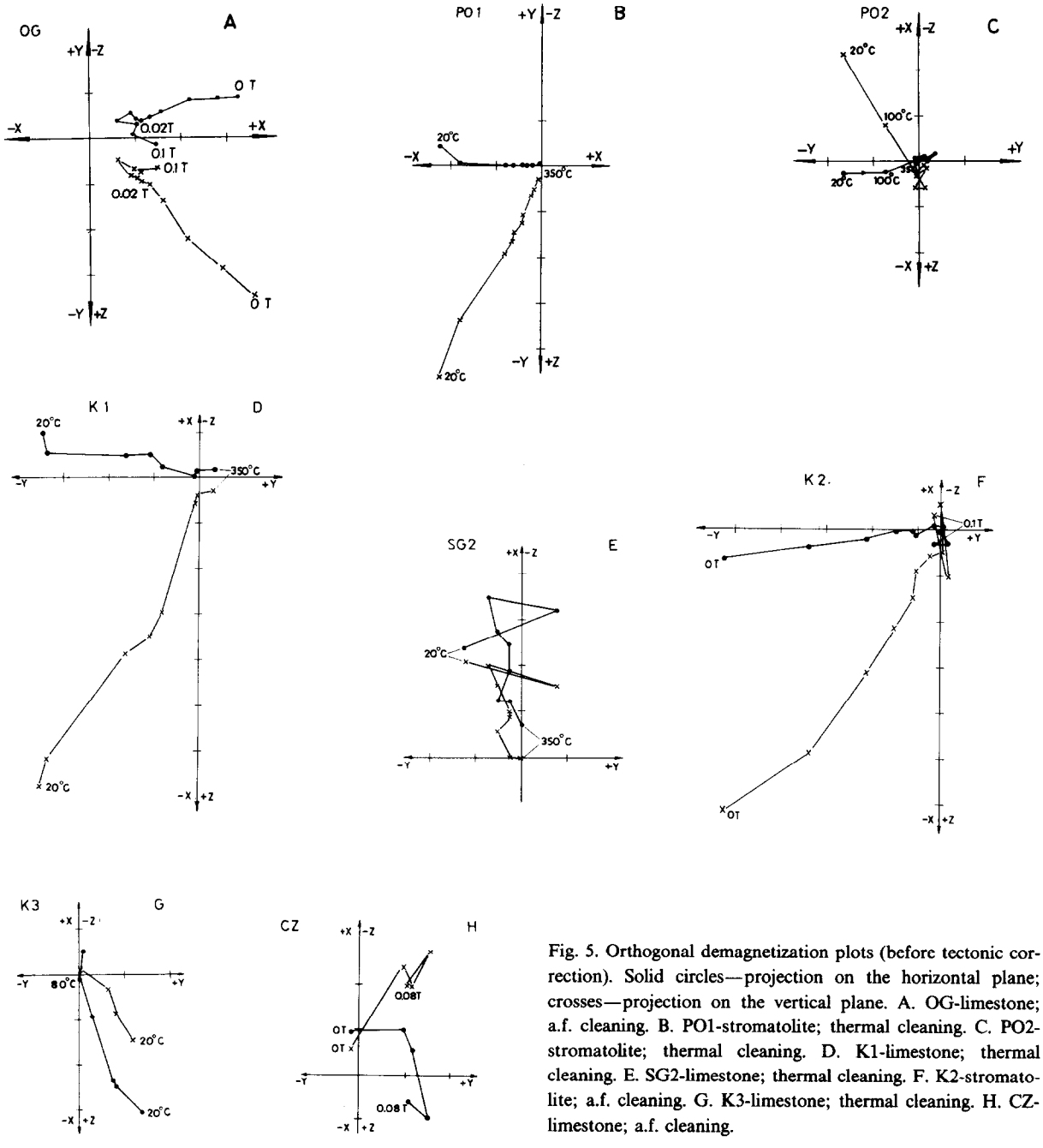



Fig. 5. Orthogonal demagnetization plots (before tectonic correction). Solid circles—projection on the horizontal plane; crosses—projection on the vertical plane. A. OG-limestone; a.f. cleaning. B. PO1-stromatolite; thermal cleaning. C. PO2-stromatolite; thermal cleaning. D. K1-limestone; thermal cleaning. E. SG2-limestone; thermal cleaning. F. K2-stromatolite; a.f. cleaning. G. K3-limestone; thermal cleaning. H. CZ-limestone; a.f. cleaning.

*Callovian*

The Callovian collection comprised 390 specimens of different limestones and stromatolites from nine exposures (Table 1). Sediments dated as early and late Callovian possessed NRM of normal polarity. The directions of NRM of middle and middle–late Callovian deposits were of reversed polarity in four exposures and of normal polarity in one (Table 2).

The NRM intensity of early Callovian limestones from W and SG2 was of the order of $10^{-5}$–$10^{-4}$ A m$^{-1}$. The typical results of demagnetizing experiments are shown in Fig. 5E. The directions of ChRM were estimated by similar procedures as those used for equally weak Oxfordian limestones. The obtained directions are surprisingly well grouped, as shown by the low values of $\alpha_{95}$ in Table 2. The mean directions obtained for these exposures are plotted in Fig. 6 and listed together with the corresponding pole positions in Table 2. The close correlation with the mean results for stable Europe suggests that these directions reflect primary magnetization.

The late Callovian stromatolite exposures Z and K2 revealed NRM intensities of about $10^{-2}$ A m$^{-1}$. A typical result of a.f. demagnetization experiments for the K2 stromatolite is shown in Fig. 5F. The linear segments appearing on orthogonal plots were interpreted as a characteristic component of NRM. The directions of ChRM for the Z stromatolites were estimated by the same procedure. The mean directions for both late Callovian exposures are plotted in Fig. 6 and listed together with the corresponding pole positions in Table 2.

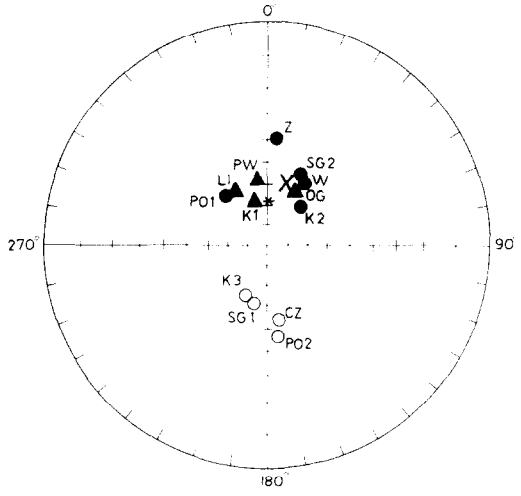



Fig. 6. Mean directions of ChRM for the exposures from the Krakow–Czestochowa Upland. Solid symbols represent normal direction. Open symbols represent reversed direction. A cross denotes the middle–late Jurassic mean direction for the stable European plate. An asterisk denotes the present local field direction. Triangles—Oxfordian; circles—Callovian. Letters denote the names of exposures referred to in the text.

TABLE 2

Mean directions of ChRM for exposures from the Krakow–Czestochowa Upland, location: $\varphi = 50.3°$ N, $\lambda = 19.5°$ E *

| Exposure | Age | $N_1/N_2$ | After demagnetization | | | | Pole position | | Polarity |
|---|---|---|---|---|---|---|---|---|---|
| | | | $D$ | $I$ | $\alpha_{95}$ | $k$ | lat. °N | long. °E | |
| OG | m. Oxfordian | 15/39 | 23.9 | 60.3 | 5.9 | 16 | 71.2 | 128.9 | N |
| LI | m. Oxfordian | 3/28 | 327.8 | 58.8 | 6.3 | 20 | 64.9 | 274.2 | N |
| PW | m. Oxfordian | 5/43 | 347.1 | 58.9 | 2.9 | 55 | 76.2 | 245.4 | N |
| K1 | e. Oxfordian | 3/60 | 337.4 | 66.2 | 3.8 | 25 | 76.3 | 288.5 | N |
| K2 | l. Callovian | 3/34 | 36.6 | 65.3 | 6.4 | 16 | 66.1 | 103.0 | N |
| Z | l. Callovian | 5/58 | 2.8 | 40.0 | 2.6 | 53 | 62.6 | 194.0 | N |
| SG1 | l. m. Callovian | 5/64 | 198.8 | −59.2 | 4.1 | 20 | 73.3 | 140.2 | R |
| CZ | m. Callovian | 2/12 | 174.4 | −55.5 | 4.6 | 91 | 75.4 | 217.9 | R |
| PO1 | m. Callovian | 3/39 | 327.9 | 59.2 | 6.3 | 14 | 65.4 | 278.0 | N |
| PO2 | m. Callovian | 3/23 | 176.6 | −46.7 | 3.7 | 68 | 67.7 | 207.5 | R |
| K3 | m. Callovian | 9/71 | 203.9 | −61.8 | 3.3 | 26 | 72.0 | 125.2 | R |
| SG2 | e. Callovian | 3/41 | 22.7 | 47.3 | 5.2 | 20 | 62.2 | 152.8 | N |
| W | e. Callovian | 11/49 | 26.4 | 52.2 | 3.7 | 32 | 63.4 | 142.1 | N |

* $N_1$—number of hand samples; $N_2$—number of specimens used in final calculations; $D$, $I$—declination, inclination (deg); $\alpha_{95}$—semi-angle of 95% confidence cone (deg.); $k$—Fisher precision parameter.

The results obtained for Z differ significantly from those for stable Europe. There is a possibility that the Z stromatolites have been remagnetized due to the presence of Fe–Mn compounds (Giżejewska and Wieczorek, 1977).

Rocks from middle Callovian exposures SG1, K3 and CZ were found to contain reversed NRM directions. In the 60 cm long section of PO stromatolite, samples from the lower part revealed a NRM of reversed polarity (PO2), while the NRM of samples from the upper part of this section had a normal polarity (PO1).

Limestones from SG1 contained reversed NRMs of intensities as low as the normally magnetized limestones from SG2, and the composition of the magnetic fraction appears to be similar (see Table 1). The mean direction of the reversed ChRM obtained by the same procedure as that for SG2 differs by about 180° in declination from the normal ChRM of SG2 (Fig. 6, Table 2).

In sandy limestones from K3 and oolitic limestones from CZ the intensities of NRMs of reversed directions were of the order of $10^{-5}$–$10^{-4}$ A $m^{-1}$. Figures 5G and 5H show orthogonal plots typical for samples from these exposures. According to rock magnetic studies (Table 1) goethite is the main magnetic mineral in both and is responsible for quick thermal demagnetization of NRM of the K3 limestones. The directions of ChRM for these limestones estimated as directions of linear segments or by the criterion of the best grouping are well grouped as shown by the low $\alpha_{95}$ values in Table 2. The mean direction obtained for K3 differs in declination by about 180° from the mean direction of the ChRM of normal polarity for the late Callovian stromatolites from K2 (see Fig. 6 and Table 2).

The results of the a.f. demagnetization experiments for CZ specimens show that the NRM is a resultant of the low- and high-coercivity components of different directions. The high-coercivity component is probably connected with goethite. According to our investigations (Kruczyk et al., 1980) the low-coercivity component is a viscous remanence acquired in the present day field. The stable component carried by goethite is interpreted as the ChRM, a primary component formed during sedimentation. The mean direction of ChRM is plotted in Fig. 6 and together with the corresponding pole position listed in Table 2. The direction differs slightly from those for K3 and SG1.

The PO stromatolite section along which the NRM direction changed from R to N will be described more thoroughly than other exposures. The rock magnetic studies revealed that both normally and reversely magnetized samples from PO contain goethite and a small amount of hematite (Table 1, Fig. 2D). The amount of hematite varies from specimen to specimen. Mineralogical studies indicate that goethite is a primary mineral occurring in various forms.

The intensities of NRM of both normally and reversely magnetized stromatolites were $10^{-2}$ A $m^{-1}$. The NRM was resistant to a.f. demagnetization. In most cases at least 40% of NRM remained after demagnetization in a field of 0.1 T. Figures 5B, C show typical orthogonal plots obtained for thermally cleaned reversed and normal specimens. The temperature required for isolating the ChRM in specimens containing only goethite were in most cases lower than 100° C, which is below the blocking temperatures of goethite (see Fig. 5C). In some specimens containing goethite and hematite higher temperatures (up to 550°) were necessary for isolating the ChRM (Fig. 5B). The directions of ChRM were estimated as directions of linear segments or stable end points. The results of rock

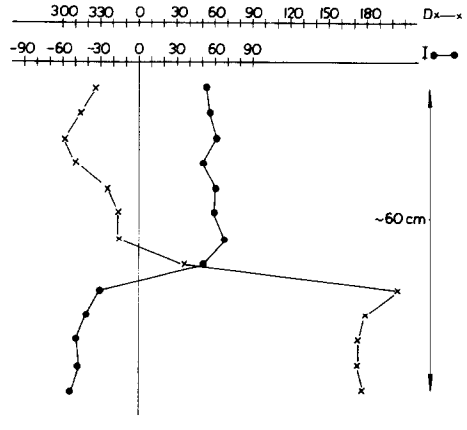



Fig. 7. Changes of declination $D$ and inclination $I$ of the ChRM determined from samples collected along the PO section. $D$ and $I$ in degrees.

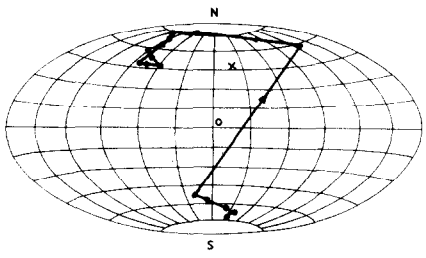



Fig. 8. Tentative path of the geomagnetic pole during the R–N transition in the PO section.

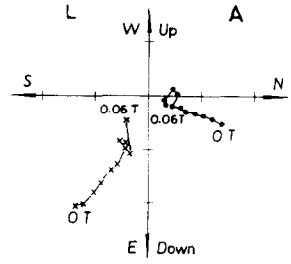
magnetic and paleomagnetic studies of PO stromatolites indicate that the carriers of ChRM isolated are primary goethite and hematite.

The change in direction from R to N of the ChRM isolated from samples collected upwards along the PO section by 180° is presented in Fig. 7 (the distances between the consecutive points are arbitrary). The directions plotted represent means for slices cut from hand samples (Kruczyk et al., 1980). The pole positions calculated from consecutive ChRM directions are plotted in Fig. 8. They form the supposed transition path of the geomagnetic pole during the examined R–N reversal. The cross in Fig. 8 indicating the sampling site lies near to the assumed path, which is in agreement with Hoffman's theory of R–N reversals of the geomagnetic field (Fuller et al., 1979).

The mean stable directions and pole positions for PO1 and PO2 calculated from the directions obtained after cleaning are plotted in Fig. 6 and listed in Table 2. The mean normal PO1 magnetization correlates well with those results from the Oxfordian rocks and is different from the results for stable Europe. The reverse PO2 data are close to the results obtained for CZ.

*Lower Subtatric Nappe*

An amount of 340 specimens of red and green radiolarites and nodular limestones from three localities CH, L and F in three different valleys was analysed paleomagnetically. The results obtained have already been reported (Kądziałko-Hofmokl et al., 1985) and are therefore summarized only briefly.



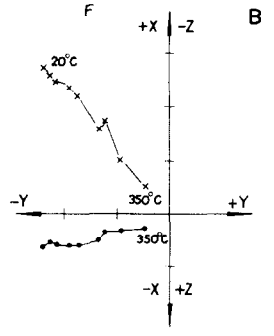




Fig. 9. Orthogonal demagnetization diagrams. A. L-limestone; a.f. demagnetization after tectonic correction. B. F-radiolarite; thermal demagnetization before tectonic correction. Symbols as in Fig. 5.

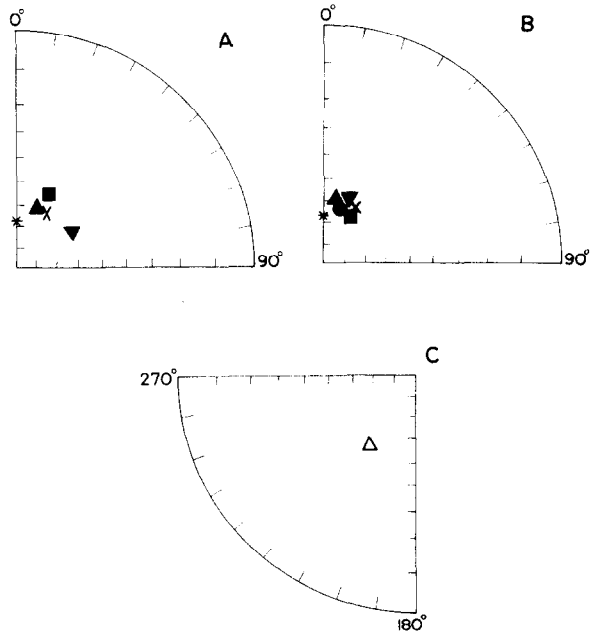



Fig. 10. Mean directions of ChRM for stratigraphic stages for the Lower Subtatric Nappe. Upturned triangle—Kimmeridgian; rectangle—Bajocian–Bathonian. Other symbols as in Fig. 6. A. CH valley. B. L valley. C. F Valley.

TABLE 3

Mean directions of ChRM for exposures from the Lower Subtatric Nappe (location: $\varphi = 49.3°$ N, $\lambda = 19.8°$ E) *

| Exposure | Age | $N_1/N_2$ | After demagnetization | | | | Pole position | | Polarity |
|---|---|---|---|---|---|---|---|---|---|
| | | | $D$ | $I$ | $\alpha_{95}$ | $k$ | lat. °N | long. °E | |
| CH | Kimmeridgian | 1/10 | 57.9 | 59.3 | 7.2 | 46 | 48.9 | 100.0 | N |
| L | Kimmeridgian | 1/26 | 17.6 | 58.4 | 2.7 | 113 | 73.8 | 142.3 | N |
| CH | Oxfordian | 3/83 | 20.1 | 60.8 | 15.1 | 67 | 74.1 | 130.3 | N |
| L | Oxfordian | 3/57 | 13.4 | 58.7 | 12.4 | 100 | 76.2 | 150.8 | N |
| F | Oxfordian | 1/24 | 216.5 | −54.7 | 7.7 | 19 | 59.9 | 123.8 | R |
| L | Callovian | 5/65 | 16.0 | 60.5 | 8.0 | 93 | 76.3 | 138.8 | N |
| CH | Bajocian–Bathonian | 1/41 | 22.4 | 53.9 | 3.9 | 33 | 67.8 | 143.5 | N |
| L | Bajocian–Bathonian | 1/34 | 29.3 | 65.3 | 4.6 | 29 | 70.6 | 104.2 | N |

* $N_1$—number of sites, $N_2$—number of specimens used in final calculations; if $N_1 \geqslant 3$, site mean has been calculated. Other symbols as in Table 2.

The intensities of the NRM for all three rock types were about $10^{-2}$–$10^{-3}$ A m$^{-1}$. The NRM stability varied; it was not possible to establish a single demagnetizing field amplitude, or a single temperature for isolating the characteristic component from the whole specimens collection. Examples of demagnetization behavior are presented in Fig. 9. The ChRM was determined from the grouping of the directions of magnetization at different demagnetization levels.

The mean directions are presented in Fig. 10. The grouping improved after tectonic corrections, indicating that the ChRM identified is of prefolding origin.

The directions of NRM and ChRM of sediments from localities CH and L were of normal polarity. Only specimens from exposure F, Oxfordian red radiolarites, possessed a reversed remanence.

Table 3 summarizes the mean directions and pole positions obtained for each sampling locality. Table 4 presents the same results grouped according to stratigraphic stages. It is clear that mean directions and pole positions obtained for the sediments from CH and L are similar and correlate well with both the data for the Krakow–Czestochowa Upland and also with the result for stable Europe.

## Conclusions

The results presented show that during the Jurassic both normal and reversed polarities of the paleomagnetic field occurred.

In the Krakow–Czestochowa Upland, the ChRM of normal direction, comparable to the European result for the middle–late Jurassic, was found in samples from localities OG, W, SG2 and K2 of late Oxfordian, early Callovian and late Callovian age, respectively. The agreement of the

TABLE 4

Stage-mean directions and pole positions for Lower Subtatric Nappe calculated after data from Table 3, without CH-Kimmeridgian *

| Age | $N_1/N_2$ | $D$ | $I$ | $\alpha_{95}$ | $k$ | Pole position | | Polarity |
|---|---|---|---|---|---|---|---|---|
| | | | | | | lat. °N | long. °E | |
| Kimmeridgian | 1/26 | 17.6 | 58.4 | 2.7 | 113 | 73.8 | 142.3 | N |
| Oxfordian | 7/64 | 19.8 | 59.2 | 6.0 | 101 | 73.2 | 135.8 | M |
| Callovian | 5/65 | 16.0 | 60.5 | 8.0 | 93 | 76.3 | 138.8 | N |
| Bajocian–Bathonian | 2/75 | 23.6 | 56.2 | 8.8 | 110 | 68.8 | 137.2 | N |

* For legend see Table 3

derived paleomagnetic pole with that for stable Europe is more than coincidental. We therefore interpret the ChRM defined to be primary and of syndepositional origin (i.e. there are no remanences of post-Jurassic origin) and to be representative of the paleomagnetic field of the middle–late Jurassic.

The Callovian belongs to the so called "Jurassic Quiet Zone" period during which the geomagnetic field was believed to be of normal direction. It is only in recent years that oceanic magnetic anomalies have been detected, suggesting the occurrence of a few reversals of the field at that period (Cande et al., 1978; Bryan et al., 1980; Gradstein and Sheridan, 1982, Channel et al., 1982). Our results obtained for four middle Callovian exposures K3, SG1, PO2 and CZ help to support the magnetic anomaly evidence.

The results obtained for the section at PO indicate that the geomagnetic field underwent reversal from R to N during the middle Callovian, as indicated by results of oceanic magnetic anomaly research.

The mean directions and pole positions obtained for reversally magnetized middle Callovian CZ limestones, normally magnetized PO1 stromatolites and PO2 stromatolites of reversed direction of NRM differ from that expected for stable Europe data. We assume that the observed ChRM was acquired during the transition polarity period.

Similarly we assume that the Oxfordian sediments from LI, PW and K1 from the Krakow–Czestochowa Upland acquired their stable remanence during a transition polarity period.

Figure 11 is an attempt to correlate our reversed NRM directions with other evidence of field reversals during the "Jurassic Quiet Zone". The middle Callovian reversal results in the PO section may be correlated with the oceanic anomaly M27 or M28 observed in the South Pacific and in the Blake–Bahama region of North Atlantic (Cande et al., 1978; Bryan et al., 1980; Gradstein and Sheridan, 1982). The reversed NRM directions found in one exposure of the Lower Subtatric Nappe (F-Oxfordian) may correspond to the short reversal recorded from one Umbrian land section (Channel et al., 1982) and/or in the Swiss–French Jura (Johnson et al., 1984).

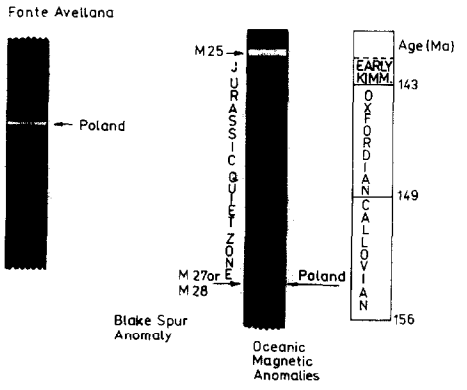



Fig. 11. The proposed correlation of Callovian and Oxfordian reversals observed in Poland with a magnetic oceanic anomalies time scale and Umbrian section (after Channel et al., 1982).

The paleomagnetic results obtained from the Lower Subtatric Nappe are better grouped than those from the Krakow–Czestochowa Upland, in spite of the more complicated tectonic situation of the former. This supports our conclusion that the NRM preserved is in fact primary and of Jurassic

TABLE 5

Mean directions and pole positions for middle–upper Jurassic *

| Region | $N$ | $D$ | $I$ | $\alpha_{95}$ | $k$ | Pole position | | Polarity |
|---|---|---|---|---|---|---|---|---|
| | | | | | | lat. °N | long. °E | |
| Krakow–Częstochowa Upland | 8 | 16.8 | 56.7 | 7.3 | 58 | 72.3 | 150.4 | M |
| Lower Subtatric Nappe | 7 | 22.3 | 59.1 | 4.3 | 198 | 71.7 | 132.2 | M |
| Balkan Bulgaria | 5 | 18.1 | 55.0 | 6.7 | 132 | 74.1 | 135.7 | N |
| Stable Europe | 12 | 15.7 | 59.0 | 5.0 | 75 | 74.7 | 147.1 | M |

* $N$—number of data used in calculations. Other symbols as in Table 2.

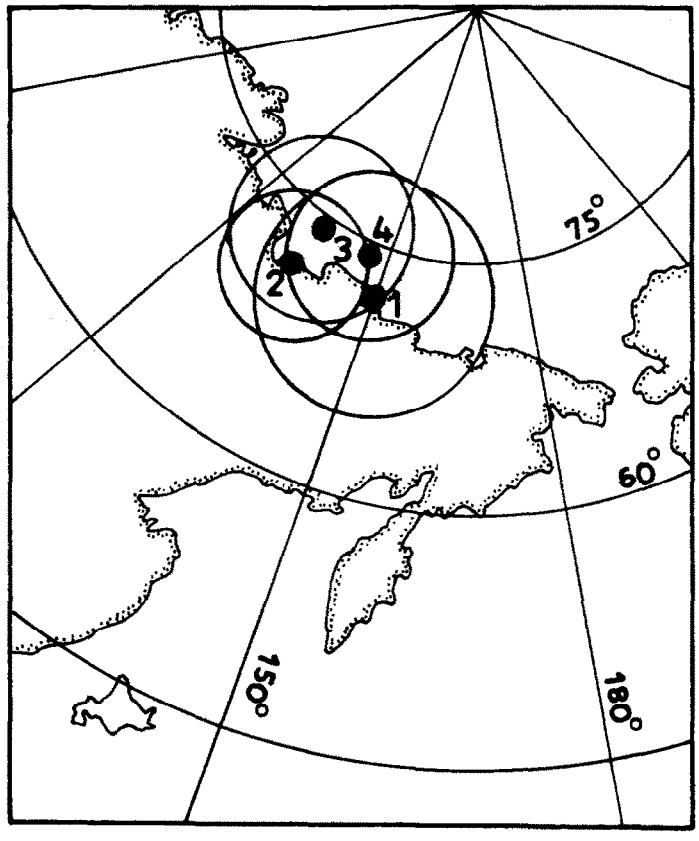



Fig. 12. Mean paleomagnetic pole positions for the middle–late Jurassic. *1*—Krakow–Czestochowa Upland, *2*—Lower Subtatric Nappe, *3*—Balkan (Bulgaria), *4*—stable European plate.

age. The slow sedimentation rate of the sediments studied will have resulted in the averaging of the short period changes of the geomagnetic field from the Bajocian to the Kimmeridgian. As can be seen from Table 3 and Fig. 10 no major rotation between the investigated areas from the three valleys has been revealed.

The comparison of results obtained for the Lower Subtatric Nappe and the Krakow–Czestochowa Upland indicates that there have been no major rotations between those two regions since the middle–late Jurassic.

The complete results are summarized in Table 5 and Fig. 12, which contain the mean directions and pole positions for the middle-late Jurassic period for the Krakow–Czestochowa Upland and Lower Subtatric Nappe, belonging to stable and mobile Europe, respectively. Also included are the previously published data for the stable European plate, and the results from the Balkan region (Kruczyk et al., 1986).

## Summary

(1) The paleomagnetic results obtained for the middle–late Jurassic for the Krakow–Czestochowa Upland agree well with the data for stable Europe and can be treated as reference data for tectonic reconstructions in the Polish Carpathians.

(2) The paleomagnetic results obtained for the three valleys of the Lower Subtatric Nappe are in good agreement with those for the Krakow–Czestochowa Upland and for stable Europe. There were no major rotations detected paleomagnetically neither between the investigated Nappe areas nor between the lower Subtatric Nappe and stable Europe.

(3) In the middle Callovian sediments of the Krakow–Czestochowa Upland a field reversal was observed. It may be correlated with the magnetic oceanic anomalies M27 or M28.

(4) There is a possibility of a field reversal

during the Oxfordian; the observed reversed directions of ChRM in the F radiolarites from the Lower Subtatric Nappe might correspond to the reversal observed in one Umbrian section.

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